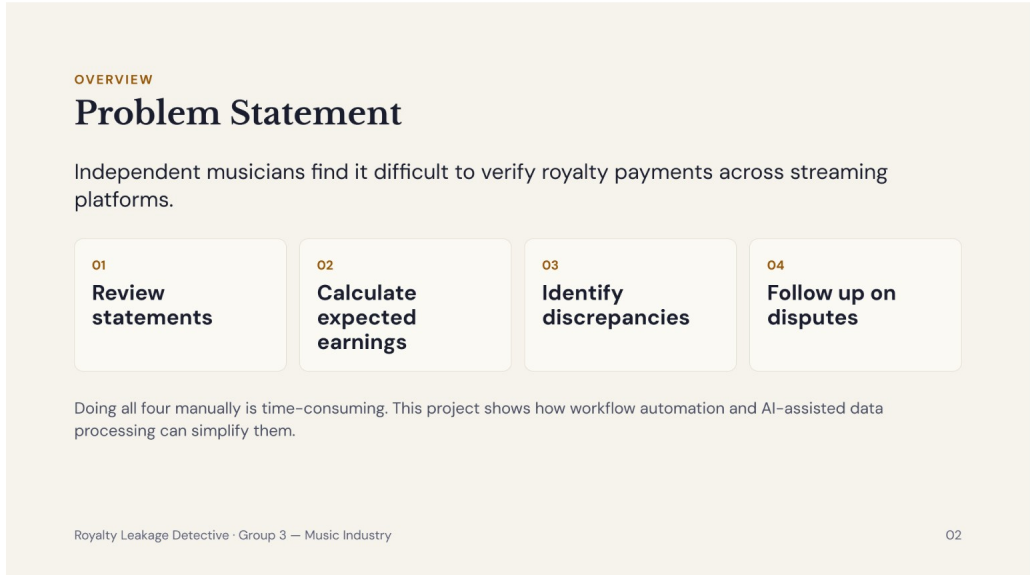
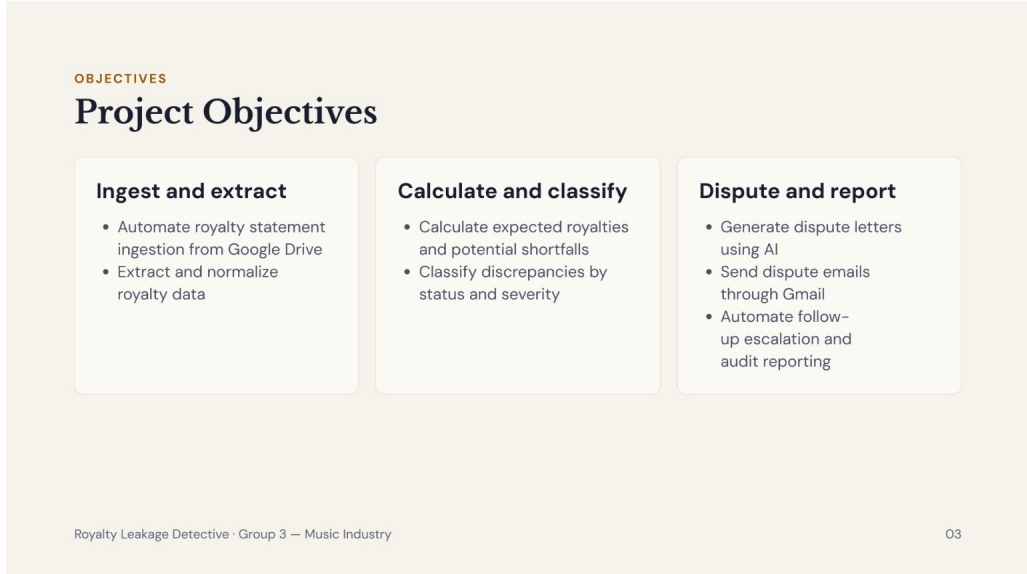


Introduce the project: an n8n workflow that reads royalty statements, calculates what should have been paid, flags gaps and drafts dispute emails.



Four manual chores make royalty verification slow: reviewing statements, calculating expected earnings, spotting discrepancies and chasing disputes. The workflow automates all four.



The seven objectives fall into three stages: ingest and extract, calculate and classify, then dispute and report.



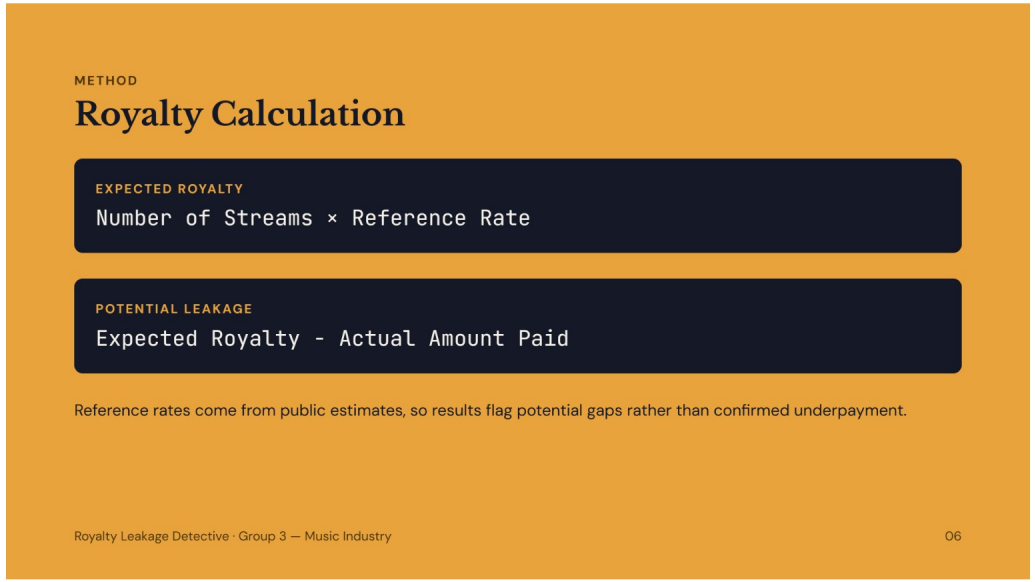
Walk through the ten steps: detection and download, extraction and AI normalization, calculation and the dispute decision, then the dispute email, the wait, escalation and the final audit summary.

TECHNOLOGY

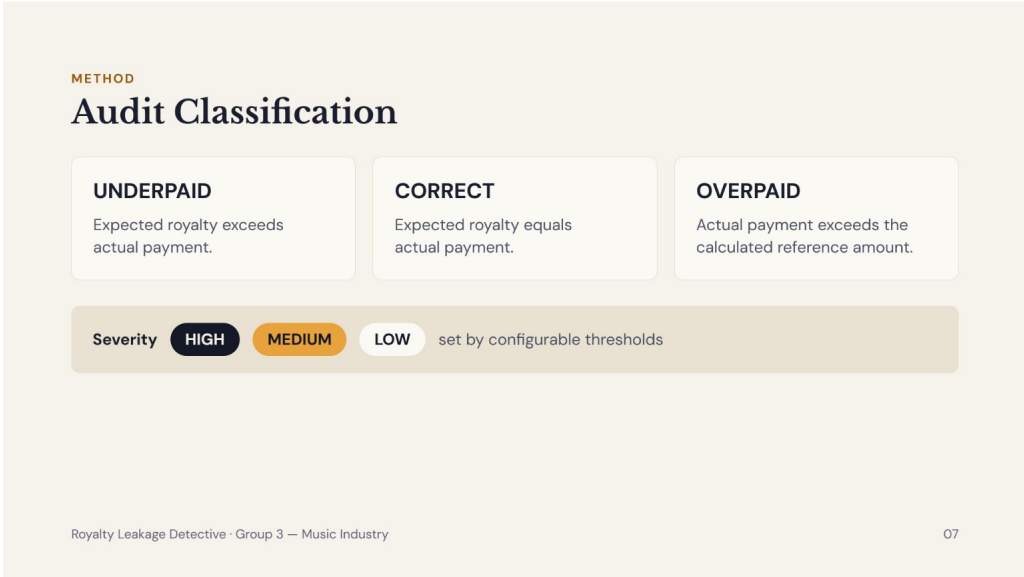
Technologies Used

Component	Role in the workflow
n8n	Workflow automation and orchestration
Google Drive	Royalty statement storage and ingestion
Extract From File	CSV data extraction
AI Agent	Data normalization and dispute-letter drafting
HTTP Request	Retrieval of public reference-rate information
JavaScript Code Nodes	Royalty calculations and audit classification
IF Node	Conditional routing
Gmail	Dispute and escalation emails

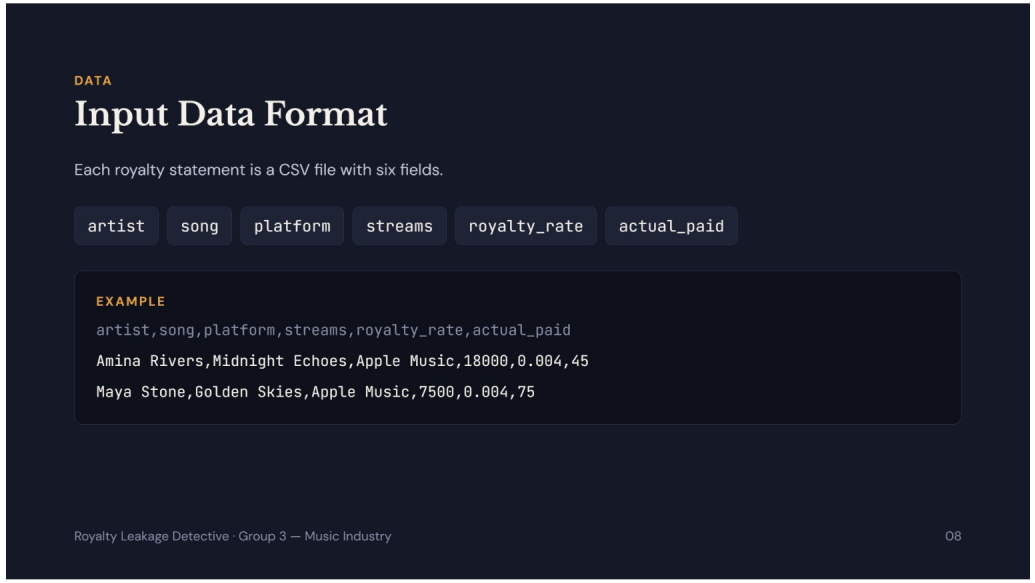
Eight building blocks: n8n orchestrates, Google Drive supplies statements, the AI Agent normalizes data and drafts letters, code nodes calculate, the IF node routes and Gmail delivers.



Two formulas drive the audit. Expected royalty is streams times the reference rate. Potential leakage is expected royalty minus what was actually paid.



Each statement row gets a status of underpaid, correct or overpaid, and discrepancies are graded high, medium or low using thresholds the user can configure.



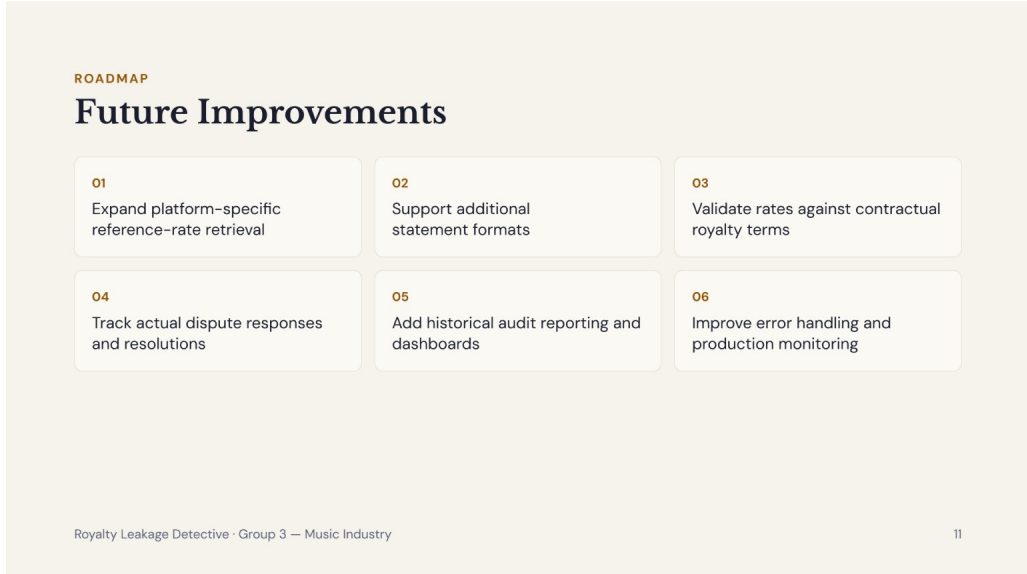
The workflow expects a simple CSV with six columns. The sample rows are the ones used to test the audit logic.



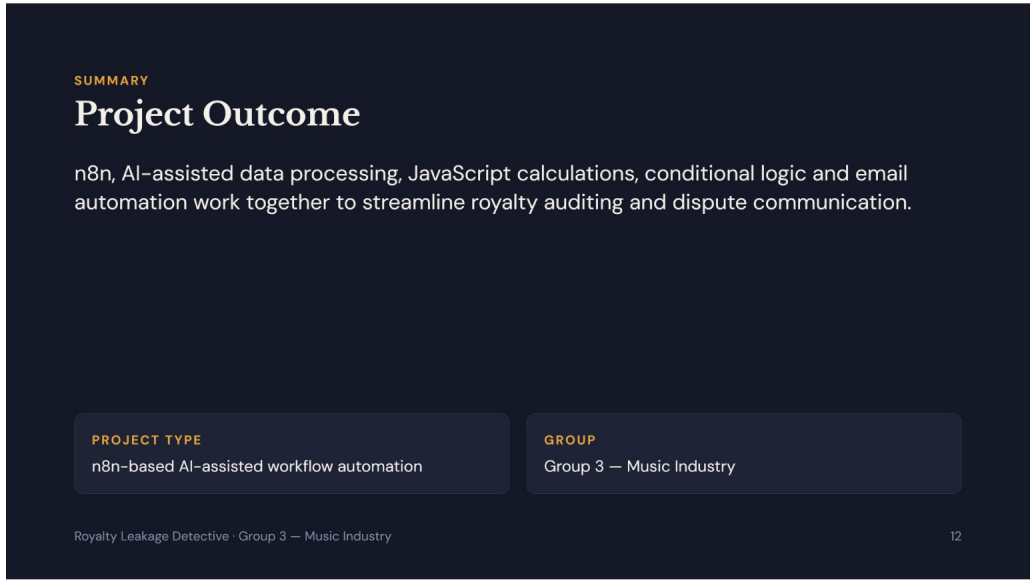
Applying the formulas to the two sample rows: the first artist is paid 27 less than the reference amount, the second is paid 45 more. Amounts are in the statement's own currency units.



Be upfront: reference rates are public estimates and the output is a list of potential discrepancies for a human to review, not proof of underpayment.



Six next steps: broader rate retrieval, more statement formats, contract-based validation, response tracking, historical dashboards and production-grade monitoring.



Close by restating the outcome: a working prototype that ties ingestion, AI normalization, calculation, routing and email into one auditable workflow.